

Mapping Propeller Noise Across Scales

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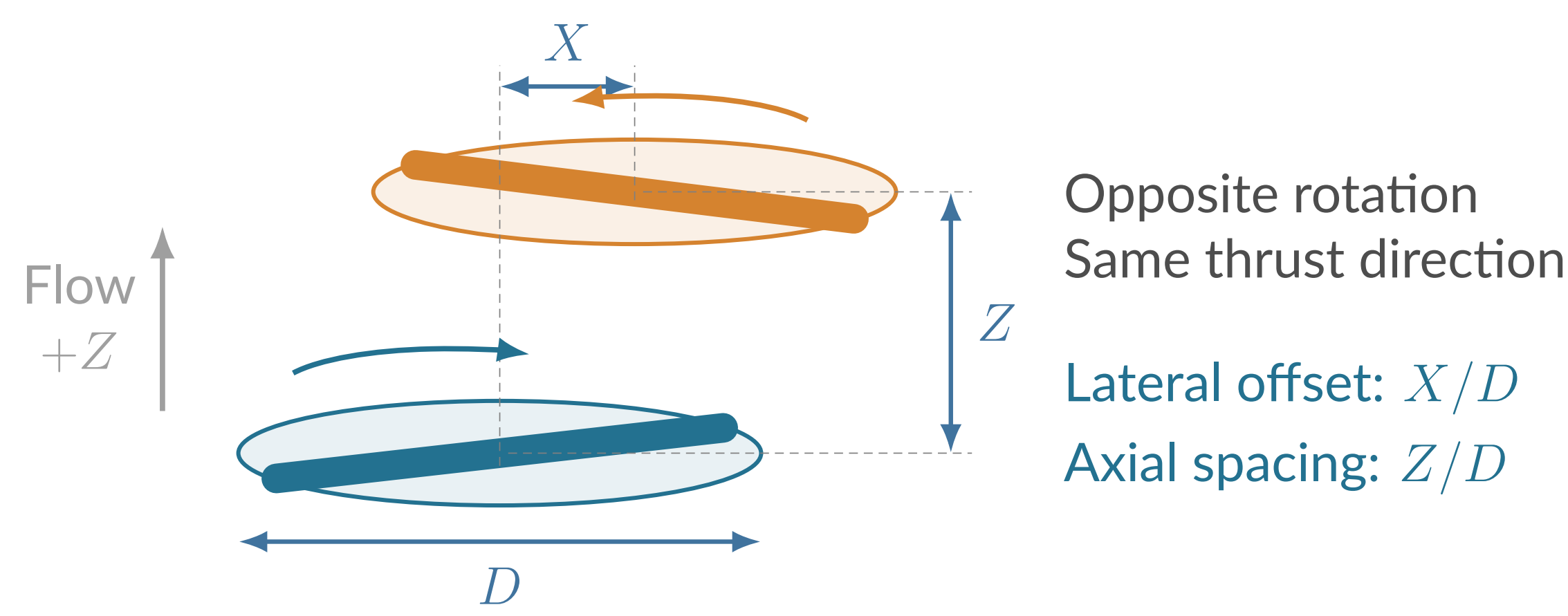
Abstract

Can small propeller tests reveal noise patterns at larger scales? We combine a custom test platform, measurements at three diameters and CFD to study spacing dependent noise. Initial similarities motivate a mapping that could reduce the cost of evaluating rotor layouts.

Introduction: a wider design space

Multirotor aircraft offer a broad design space, from side by side propellers to staggered and vertically offset layouts. [1,2] Rotor placement can improve aerodynamic efficiency, but benefits depend on configuration and operating conditions. [1] Layout also changes wake interactions and radiated noise. [2,3] We aim to learn a transferable noise mapping from smaller counter rotating propellers, enabling **efficient screening of candidate layouts** before larger experiments.

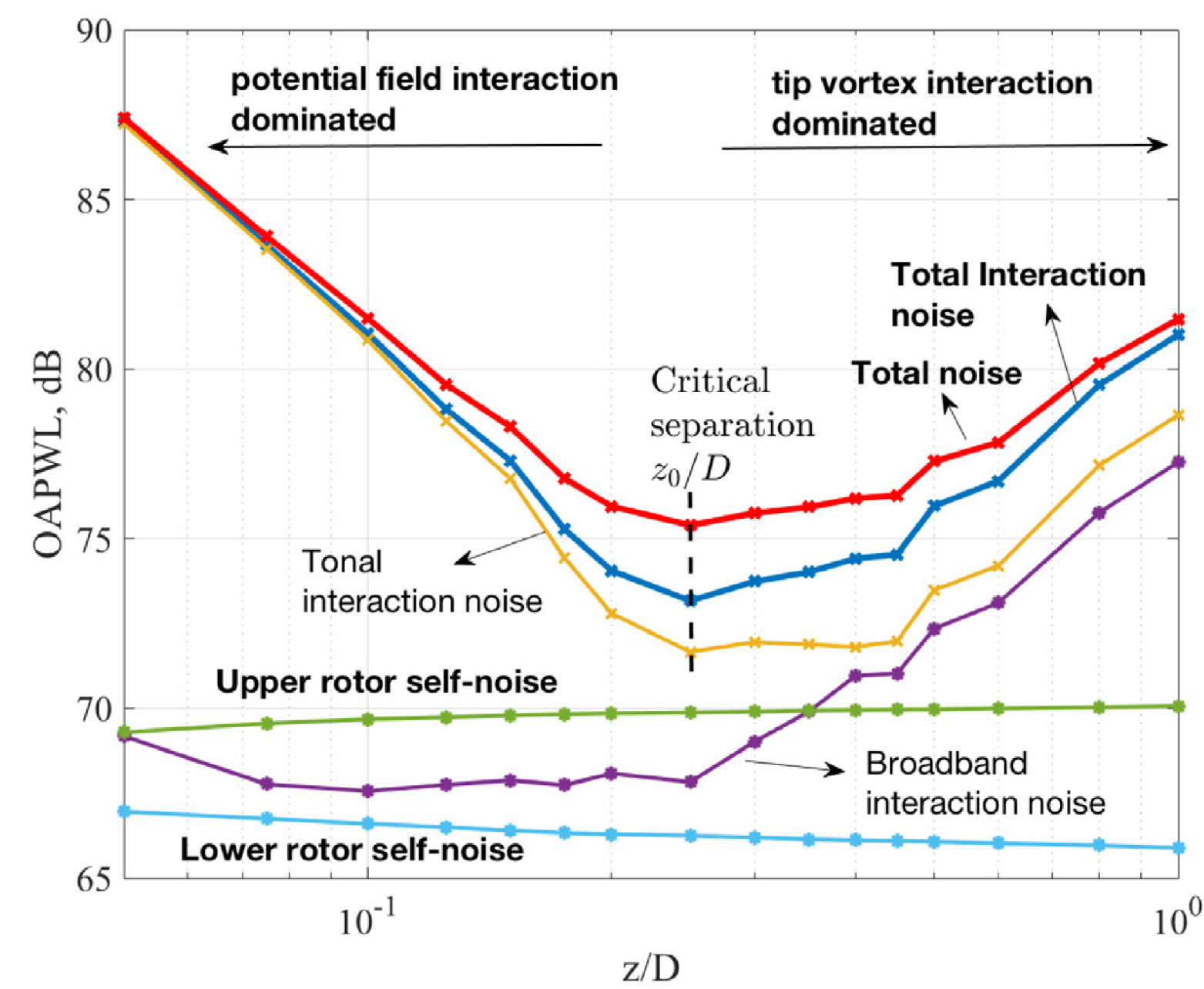
The geometry of the question



X : lateral offset. Z : axial spacing. D : propeller diameter.

Changing X/D and Z/D changes how the rotors interact. We ask which spacing dependent noise patterns persist when the propellers are scaled.

Spacing changes the noise mechanisms



Published sound power trends at 16 N total thrust. Chaitanya and colleagues in [4], Fig. 14, CC BY 4.0.

Close spacing

Potential interaction produces unsteady loading.

Wake encounters

Velocity deficits and tip vortices disturb the other rotor.

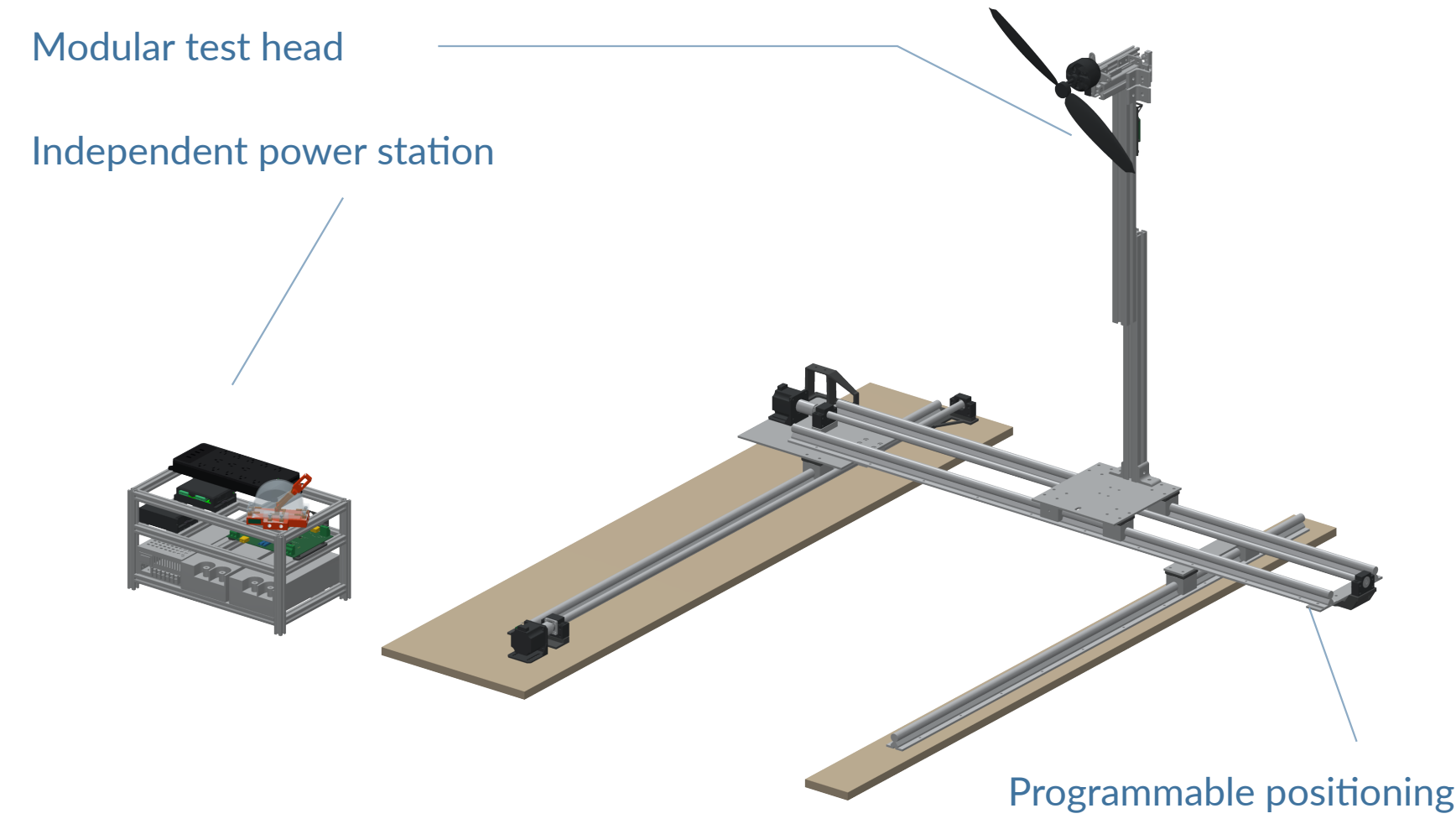
Across scales

Do these contributions change in a predictable way?

References

- [1] Kostek et al. (2026). Influence of vertical rotor spacing on aerodynamic efficiency of quadcopter configurations.
- [2] Granata et al. (2024). Aerodynamic and aeroacoustics investigation of tandem propellers in hover for eVTOL configurations.
- [3] Kingan and Parry (2020). Time domain analysis of contra rotating propeller noise: wake interaction with a downstream propeller blade.
- [4] Schram and Bennett (2024). Aeroacoustics research in Europe: CEAS ASC report on 2022 highlights.
- [5] Ffowcs Williams and Hawkings (1969). Sound generation by turbulence and surfaces in arbitrary motion.

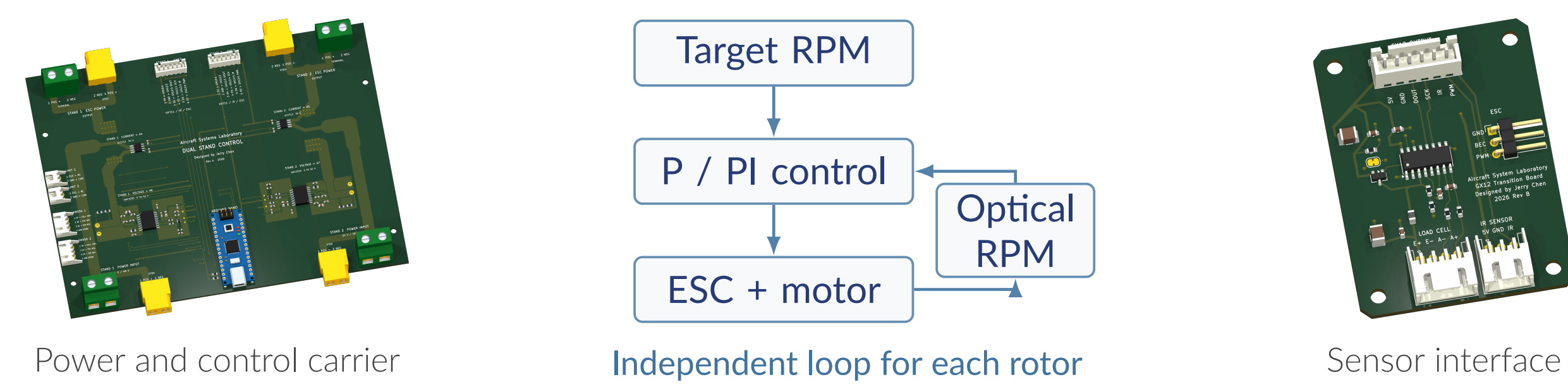
A platform developed from the ground up



Project CAD. Motor and 0.5 m propeller are illustrative.

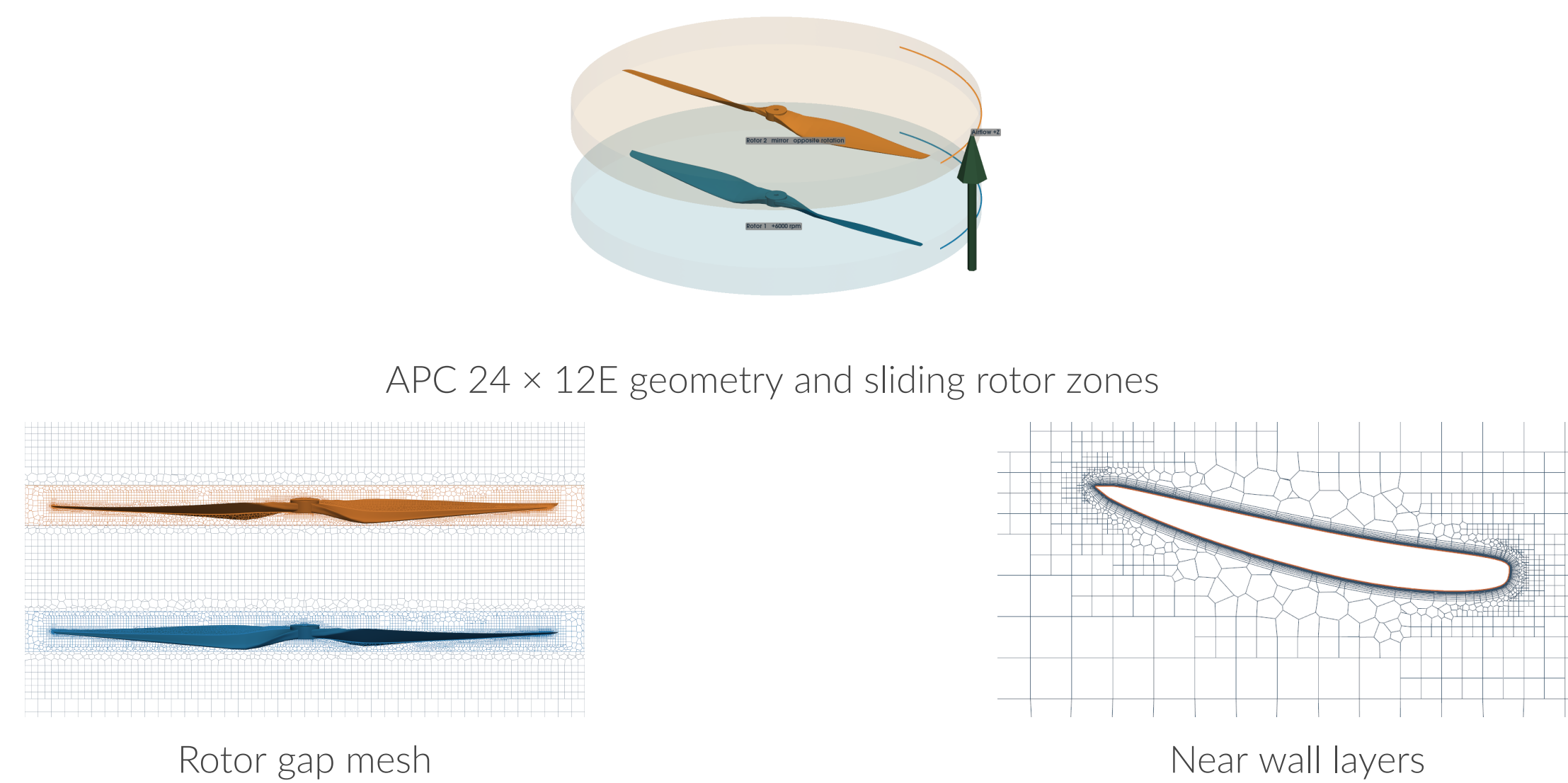
We developed the mechanics, electronics and experiment software. Adjustable geometry enables spacing sweeps; independent RPM control supports repeatable comparisons.

Custom electronics and control



Two custom PCBs consolidate power, sensing and command wiring. Independent P / PI loops regulate rotor speeds; Python schedules tests and logs paired telemetry.

CFD and acoustic prediction



Pilot: $D = 0.6096$ m; ± 6000 rpm; $X/D = 0$, $Z/D = 0.25$.

Flow: SST $k\omega$ URANS; sliding mesh; $\Delta t = 27.78 \mu s$.

Initial mesh: 7.39 million cells; 18 prism layers specified; $5 \mu m$ first layer.

$$p'_{FWH} \approx \underbrace{p'_T}_{\text{thickness}} + \underbrace{p'_L}_{\text{loading}}$$

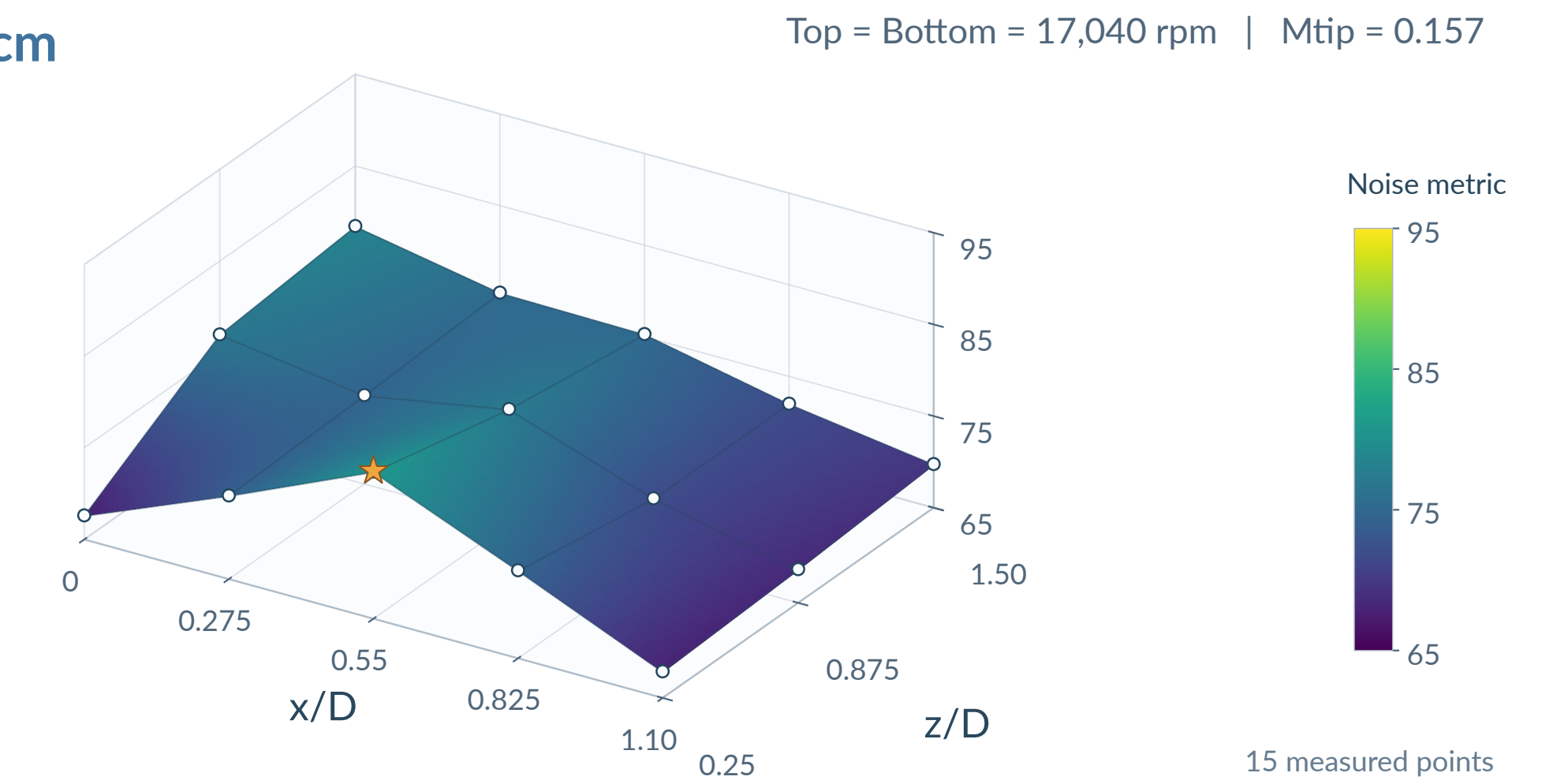
Blade and hub wall sources predict resolved periodic sound. [5] Exterior volume radiation is omitted. CFD acoustics across operating conditions remain in progress.

Experimental validation. Compare measured and predicted noise trends at matched spacing and speed. Mesh and time step checks assess numerical sensitivity; repeated measurements will establish repeatability and **validate the numerical methods**.

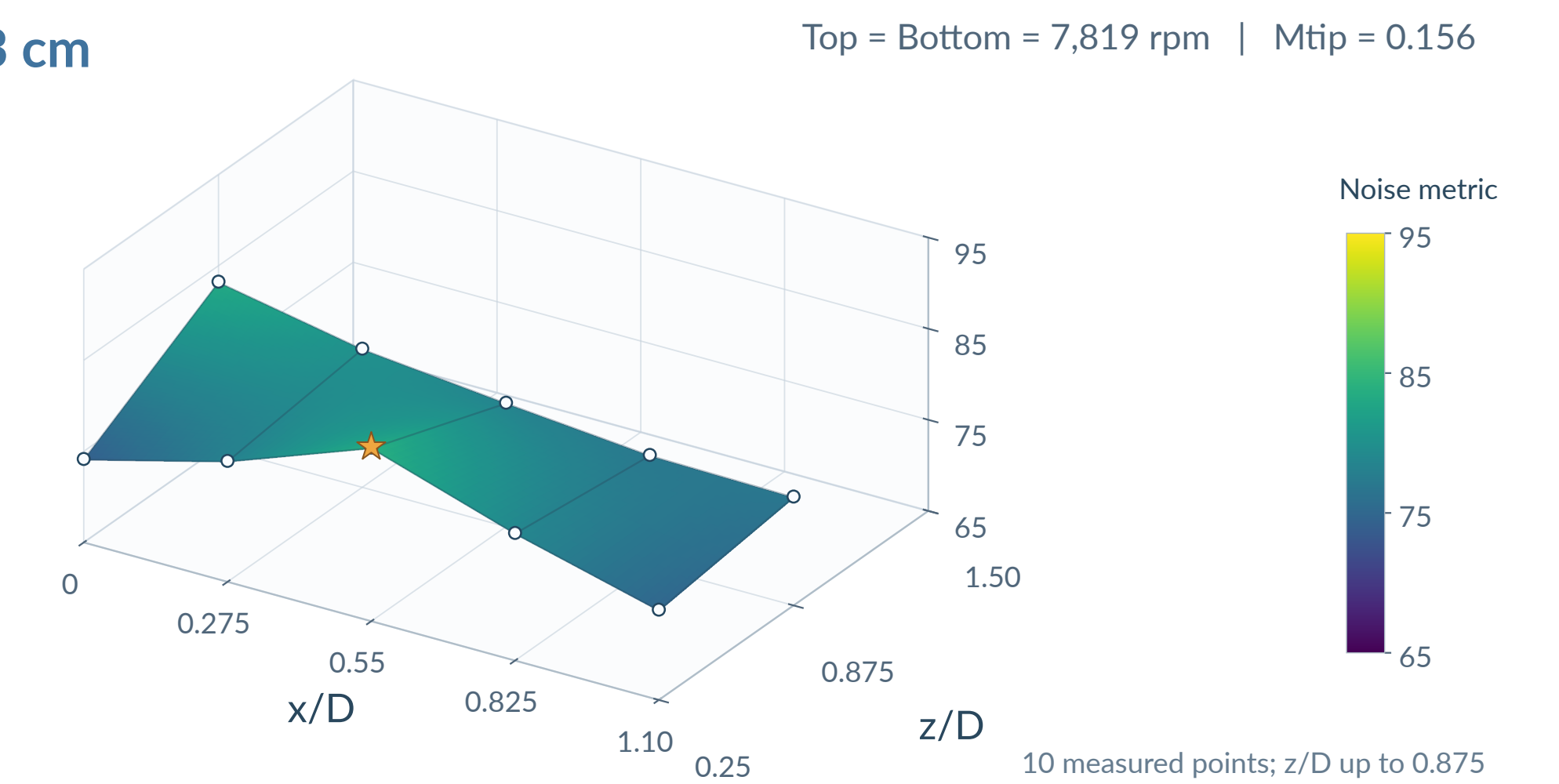
Preliminary measurements across three scales

One paired RPM condition per diameter, selected at **nearly matched tip Mach number**: $M_{tip} \approx 0.157$ for both rotors, using $a = 340.3$ m/s.

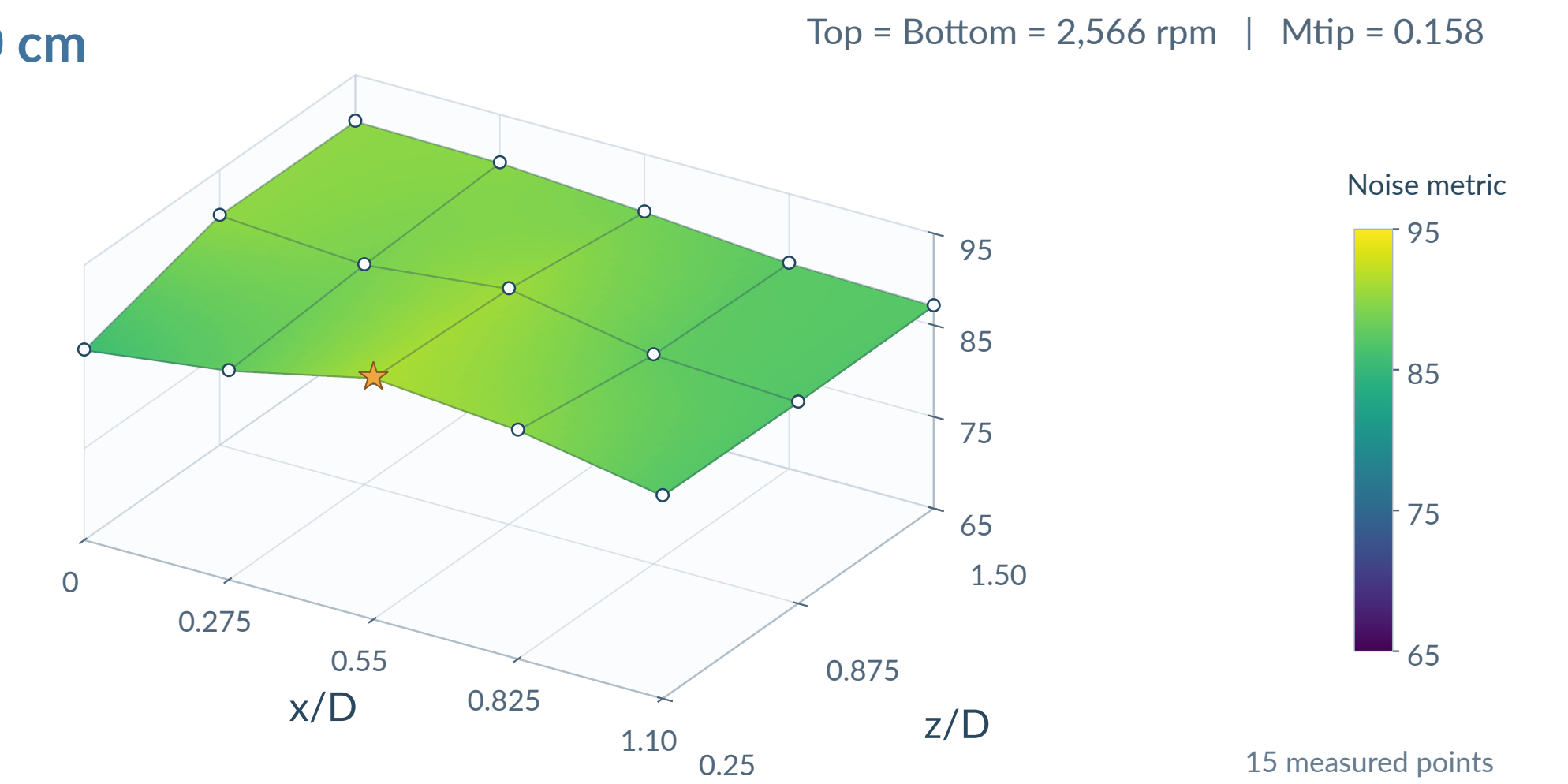
a $D = 6$ cm



b $D = 13$ cm



c $D = 40$ cm



Dots: measurements; stars: sampled maxima. Surfaces interpolate measured coverage only.

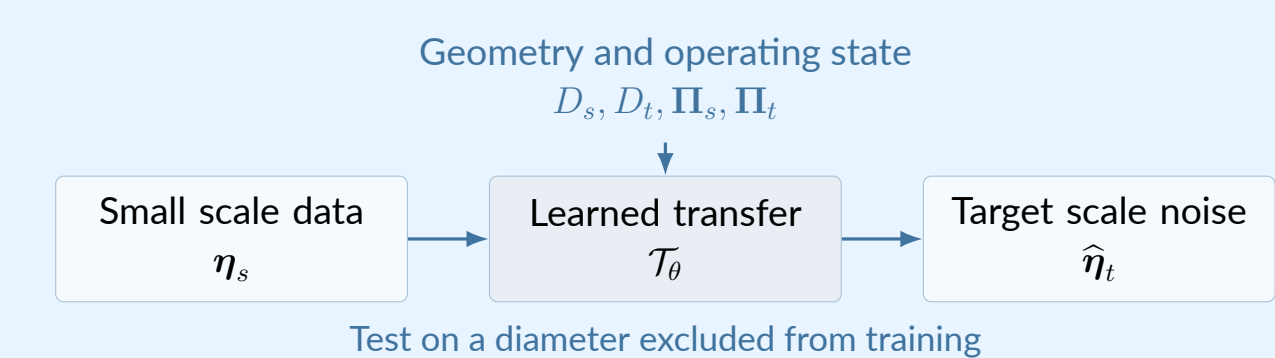
RMS of reported microphone summaries; units unconfirmed. The 13 cm data end at $z/D = 0.875$.

At $z/D = 0.25$, all three scales share a measured maximum at $x/D = 0.55$. Pairwise Spearman rank correlations are 0.89 to 0.96 across ten common positions.

Our hypothesis: mapping across diameters

Working hypothesis. Geometry and operating state may help transfer spacing trends, even when absolute noise levels differ.

$$\hat{\eta}_t = \mathcal{T}_\theta(D_s, D_t, \Pi_s, \Pi_t)[\eta_s]$$



η : noise pattern over spacing and frequency. Π : geometry and operating conditions, including Mach and Reynolds numbers. **Next test:** fit a mapping and predict measurements at a diameter excluded from training.